REQUIREMENTS ELICITATION DOCUMENT

Restaurant Ordering System

Menus, Orders, Payments, and Delivery Management

GROUP 8

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INTRODUCTION

Project Background & Problem Statement

Many restaurants still use manual methods for taking orders and managing deliveries.

Customers often experience delays when ordering food through phone calls or in-person visits.

Restaurant staff spend significant time managing menus, orders, and payments manually.

Food delivery tracking can be inefficient, resulting in delays and customer dissatisfaction.

Problem statement: The lack of a centralized ordering system causes delays, order mistakes, payment issues, and poor customer experience.

Proposed solution: A Restaurant Ordering System that centralizes menu management, order processing, payments, and delivery tracking.

VISION

Business Goals & Objectives

Improve customer experience through fast and easy ordering.

Reduce order processing time and minimize order-entry mistakes.

Enable secure online payment methods.

Improve delivery management and order tracking.

Increase restaurant efficiency and sales.

Provide real-time updates to customers regarding order status.

PREPARING FOR ELICITATION

Stakeholder Identification & Analysis

Stakeholder	Role / Interest	Influence
Customer	Browse menu, place orders, make payments, track orders	High
Restaurant Staff	Process orders and manage food preparation	High
Delivery Personnel	Receive assigned deliveries and update delivery status	Medium
Restaurant Manager	Manage menu, reports, prices, and operations	High
System Administrator	Manage users, permissions, and system settings	Medium
Payment Service Provider	Process and confirm online payments	Low

PREPARING FOR ELICITATION

Stakeholder Map

Primary Stakeholders

Customers
Restaurant Staff

Secondary Stakeholders

Restaurant Manager
System Administrator

External Stakeholders

Payment Gateway
Delivery Personnel

Restaurant Ordering System

Central platform connecting customers, restaurant operations, delivery, and payment services.

PREPARING FOR ELICITATION

Requirements Elicitation Plan

Technique	Applied To	Purpose
Interviews	Restaurant Managers	Understand restaurant workflow, operational challenges, and business needs
Questionnaires	Customers	Gather customer ordering preferences and usability requirements
Observation	Restaurant Operations	Analyze current ordering, preparation, payment, and delivery processes
Brainstorming	Project Team	Generate system features and possible improvements
Use Cases	Customers, Staff, Managers	Define system behavior and interactions for key activities
Prototyping	Users / Staff	Validate screen designs and user flow before development

REQUIREMENTS

Functional Requirements

FR1 – The system shall allow customers to register and log in.

FR2 – The system shall allow customers to browse restaurant menus and view item details.

FR3 – The system shall allow customers to add menu items to a shopping cart and change quantities.

FR4 – The system shall allow customers to place orders and receive an order confirmation.

FR5 – The system shall allow customers to pay online or choose cash payment.

FR6 – The system shall allow restaurant staff to view and manage incoming orders.

FR7 – The system shall allow managers to add, edit, and remove menu items and prices.

FR8 – The system shall allow delivery personnel to view assigned deliveries and update delivery status.

FR9 – The system shall send notifications about order and delivery progress.

FR10 – The system shall provide customers with access to their order history.

REQUIREMENTS

Non-Functional Requirements

Requirement	Specification
Performance	The system should process and confirm orders within 3 seconds under normal load.
Scalability	The system should support at least 3,000 concurrent users.
Availability	The system should provide 99.5% uptime, excluding scheduled maintenance.
Security	Payment and customer data shall be protected using encryption and secure authentication.
Usability	The system should be responsive and usable on mobile and desktop devices.
Reliability	The system shall prevent duplicate order processing and preserve order data accurately.

REQUIREMENTS

Business Rules

Customers must create an account before placing orders.

Payment must be confirmed before order preparation begins for online payments.

Orders cannot be canceled once they are marked “Preparing.”

Menu prices can only be modified by authorized restaurant managers.

Delivery personnel can only view and update deliveries assigned to them.

REQUIREMENTS

System Constraints & Assumptions

Constraints

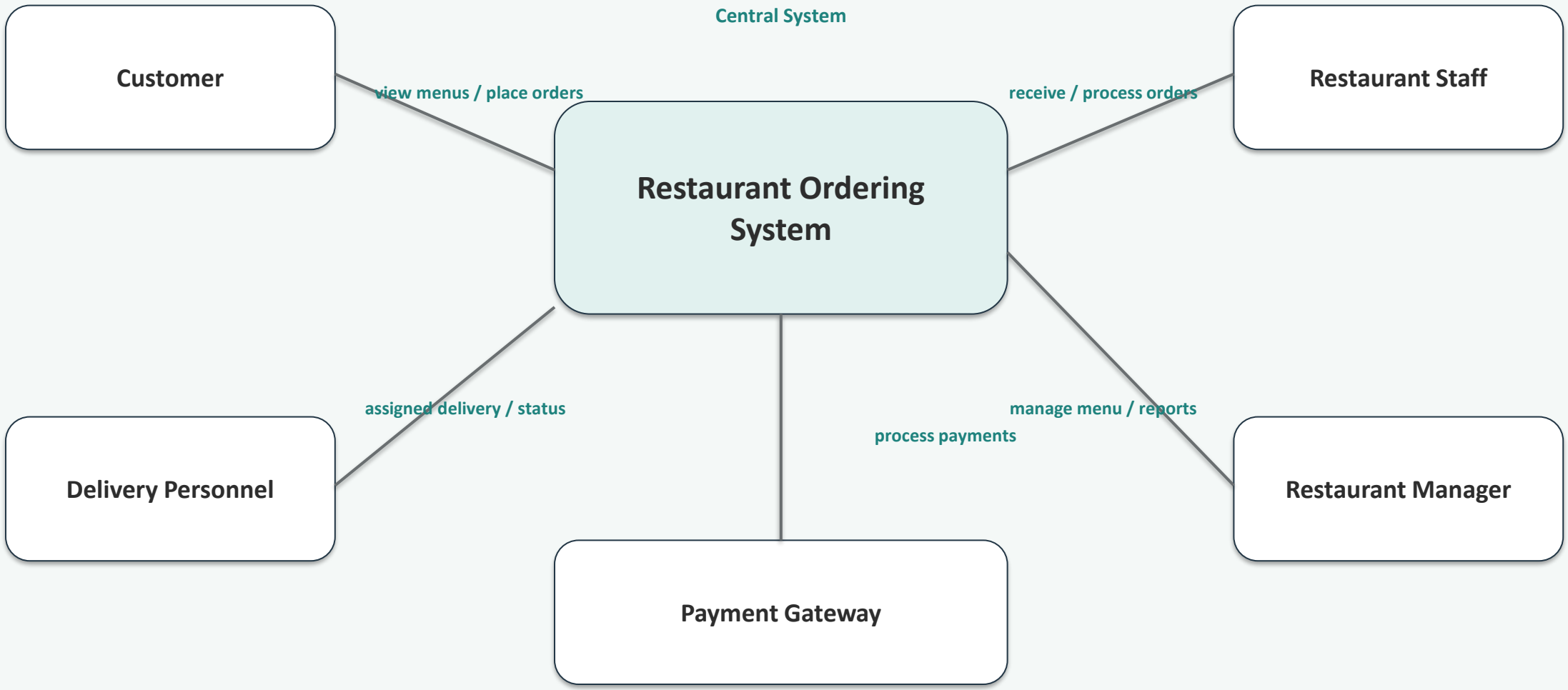
- Must be a web-based system.
- Must support mobile devices.
- Must integrate with payment gateways.
- Must support internet-based delivery tracking.
- System operation depends on reliable network connectivity.

Assumptions

- Customers have internet access.
- Delivery personnel use smartphones.
- Payment services remain available.
- Restaurant staff are trained to use the system.
- Menu and delivery information is maintained accurately.

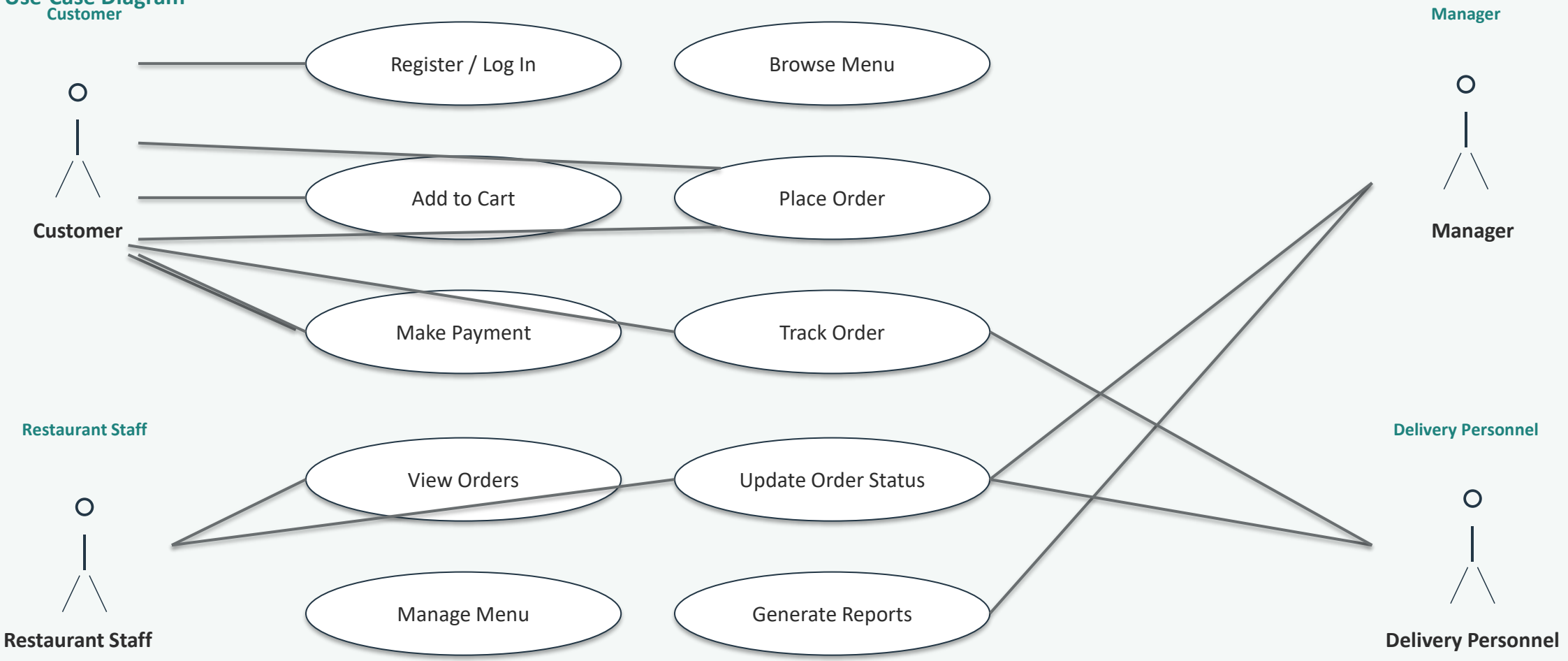
SYSTEM MODELS

Context Diagram



SYSTEM MODELS

Use-Case Diagram



SYSTEM MODELS

Use-Case Flow – Place Order

Field	Description
Primary actor	Customer
Precondition	Customer is logged in and selected available menu items.
Postcondition	Order is created, payment is recorded when applicable, and restaurant staff are notified.

1. Customer logs in.
2. Customer browses the menu and selects food items.
3. Customer adds selected items to the cart.
4. Customer reviews the cart and confirms the order.
5. Customer selects online payment or cash payment.
6. System records the order and payment result.
7. System notifies restaurant staff that a new order has been received.

Extension: Payment fails → System displays an error and requests another payment method.

ANALYSIS

Requirements Prioritization (MoSCoW)

Must Have

User registration
Menu management
Shopping cart
Order placement
Payment processing

Should Have

Delivery tracking
Notifications
Order history
Order status updates

Could Have

Customer reviews
Discount coupons
Favorite orders
Loyalty features

Won't Have (This Phase)

AI food recommendations
Voice ordering
Multi-restaurant support

WIREFRAMES · 1 OF 5

Customer – Home / Menu

Restaurant Header

Restaurant name • Profile / Login • Cart

Search Bar

🔍 Search for food or menu items

Food Card 1

Image
Name • Price
[Add to Cart]

Food Card 2

Image
Name • Price
[Add to Cart]

Food Card 3

Image
Name • Price
[Add to Cart]

WIREFRAMES · 2 OF 5

Customer – Cart & Checkout

Selected Items

Food item 1 — Quantity controls — Price
Food item 2 — Quantity controls — Price
Food item 3 — Quantity controls — Price

Delivery address
Order notes

Checkout

Subtotal
Delivery fee
TOTAL AMOUNT

Payment Options
○ Online Payment
○ Cash Payment

[PLACE ORDER]

WIREFRAMES · 3 OF 5

Restaurant Staff Dashboard

New Orders

Order #101
Order #102
Order #103

[View Order]
[Accept Order]

Preparing Orders

Order #098
Order #099

[Mark Ready]
[Update Status]

Completed Orders

Order #095
Order #096
Order #097

[View Details]

WIREFRAMES · 4 OF 5

Delivery Dashboard

Assigned Deliveries

Order #101
Customer: Name
Address: Kigali

[View Details]

Delivery Status

Order #101
○ Assigned
○ Picked Up
○ On the Way
○ Delivered

[UPDATE STATUS]

Delivery Information

Customer address
Phone/contact
Order summary
Estimated delivery time

[Contact Customer]

WIREFRAMES · 5 OF 5

Administrator Dashboard

User Management

Customers
Restaurant Staff
Managers
Delivery Personnel

[Manage Users]

Menu Management

Menu items
Categories
Prices
Availability

[Manage Menu]

Order Reports

Daily orders
Sales reports
Delivery reports
Payment reports

[View Reports]

System Settings

Payment settings
Notifications
User permissions
System configuration